Article Title

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Abstract

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1. Derivation of the SDP Safe Approximation for DRCC

1.1. Problem Formulation

We consider the Distributionally Robust Chance Constraint (DRCC) defined over a moment-based ambiguity set $\mathcal{D}_1$ with non-negative support. The ambiguity set contains all probability distributions P supported on $\mathbb{R}_+^N$ that match the empirical mean $\boldsymbol{\mu}$ and the second-order moment matrix $\mathbf{Q}$:

$$\mathcal{D}_1 = \left\{ P \in \mathcal{P}(\mathbb{R}_+^N) : \mathbb{E}_P[\mathbf{d}] = \boldsymbol{\mu},\ \mathbb{E}_P[\mathbf{d}\mathbf{d}^\top] = \mathbf{Q} \right\}, \tag{1}$$

where $\mathbf{Q} = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top$. The chance constraint requires the linear inequality $\mathbf{d}^\top \mathbf{v} \leq 0$ to hold with probability at least $1 - \gamma$. This is mathematically equivalent to constraining the worst-case violation probability to be at most γ:

$$\sup_{P \in \mathcal{D}_1} P\left(\mathbf{d}^\top \mathbf{v} > 0\right) \leq \gamma. \tag{2}$$

1.2. Dual Reformulation via Generalized Moment Theory

The primal problem is a generalized moment problem where we maximize the expectation of the indicator function $f(\mathbf{d}) = \mathbb{I}(\mathbf{d}^\top \mathbf{v} > 0)$. Assuming the Slater condition holds (i.e., $\mathbf{Q} \succ 0$), strong duality allows us to reformulate the problem as finding the tightest polynomial upper bound on $f(\mathbf{d})$ over the support $\mathbb{R}^N_+$:

$$\min_{y_0,\mathbf{y},\mathbf{M}} \quad y_0 + \boldsymbol{\mu}^\top \mathbf{y} + \mathbf{Q} \bullet \mathbf{M} \tag{3a}$$

$$\text{s.t.} \quad y_0 + \mathbf{y}^\top \mathbf{d} + \mathbf{d}^\top \mathbf{M}\mathbf{d} \geq \mathbb{I}(\mathbf{d}^\top \mathbf{v} > 0), \quad \forall \mathbf{d} \in \mathbb{R}^N_+. \tag{3b}$$

To express the quadratic polynomial in a compact form, we introduce the augmented vector $\hat{\mathbf{d}} = (1, \mathbf{d}^\top)^\top$ and the aggregated dual matrix $\mathbf{Z} \in \mathbb{S}^{N+1}$:

$$\mathbf{Z} = \begin{pmatrix} y_0 & \frac{1}{2}\mathbf{y}^\top \\ \frac{1}{2}\mathbf{y} & \mathbf{M} \end{pmatrix}. \tag{4}$$

By homogenizing the polynomial, the non-negativity constraint over $\mathbb{R}^N_+$ is equivalent to requiring the matrix $\mathbf{Z}$ to be copositive. The dual constraints naturally split into two conditions:

1. **Global Non-negativity:** The polynomial must be non-negative everywhere on the support $\mathbb{R}^N_+$.
2. **Violation Coverage:** The polynomial must be greater than or equal to 1 whenever the constraint is violated (i.e., $\mathbf{d}^\top \mathbf{v} > 0$).

1.3. Conservative Approximation via S-Procedure

The first condition corresponds to requiring $\mathbf{Z} \in \mathcal{COP}(\mathbb{R}^{N+1}_+)$, where $\mathcal{COP}$ denotes the standard copositive cone (i.e., matrices that generate non-negative quadratic forms over the non-negative orthant).

The second condition, however, is more complex. It requires the polynomial to be non-negative specifically over the *violation cone* $\mathcal{K}_{vio}$, defined as the intersection of the non-negative orthant and the half-space where the chance constraint is violated:

$$\mathcal{K}_{vio} = \left\{ \mathbf{d} \in \mathbb{R}^N_+ \mid \mathbf{d}^\top \mathbf{v} > 0 \right\}. \tag{5}$$

Mathematically, this demands that $\hat{\mathbf{d}}^\top \mathbf{Z} \hat{\mathbf{d}} \geq 1$ for all $\mathbf{d} \in \mathcal{K}_{vio}$, or equivalently:

$$\mathbf{d}^\top \mathbf{v} > 0 \implies \hat{\mathbf{d}}^\top (\mathbf{Z} - \mathbf{E}_{11}) \hat{\mathbf{d}} \geq 0, \tag{6}$$

where $\mathbf{E}_{11} \in \mathbb{S}^{N+1}$ is a matrix with 1 at index $(1,1)$ and 0 elsewhere, representing the constant term -1 in the quadratic form.

Verifying copositivity with respect to a general polyhedral cone like $\mathcal{K}_{vio}$ is known to be co-NP-complete. To obtain a computationally tractable formulation, we resort to a conservative approximation using the *S-procedure* (Lagrangian relaxation). The S-procedure allows us to replace the conditional implication (6) with a single unconditional constraint by introducing a Lagrange multiplier to penalize the violation condition.

First, we lift the linear violation condition $\mathbf{d}^\top \mathbf{v} > 0$ into the quadratic space of the augmented vector $\hat{\mathbf{d}}$. We define the matrix $\mathbf{V}_{aug} \in \mathbb{S}^{N+1}$ as:

$$\mathbf{V}_{aug} = \begin{pmatrix} 0 & \frac{1}{2}\mathbf{v}^\top \\ \frac{1}{2}\mathbf{v} & \mathbf{0} \end{pmatrix}, \tag{7}$$

such that $\hat{\mathbf{d}}^\top \mathbf{V}_{aug} \hat{\mathbf{d}} = 1 \cdot (\mathbf{v}^\top \mathbf{d}) = \mathbf{d}^\top \mathbf{v}$.

By introducing a non-negative scalar multiplier $\lambda \geq 0$, we can construct a sufficient condition for (6). If we ensure that the Lagrangian function is non-negative over the entire non-negative orthant, the original condition will hold on the subset $\mathcal{K}_{vio}$. Specifically, we require:

$$\hat{\mathbf{d}}^\top (\mathbf{Z} - \mathbf{E}_{11}) \hat{\mathbf{d}} - \lambda (\hat{\mathbf{d}}^\top \mathbf{V}_{aug} \hat{\mathbf{d}}) \geq 0, \quad \forall \mathbf{d} \in \mathbb{R}^N_+. \tag{8}$$

Rearranging the terms, this condition is equivalent to requiring the aggregated matrix to be copositive with respect to the standard cone:

$$\hat{\mathbf{d}}^\top \left(\mathbf{Z} - \mathbf{E}_{11} - \lambda \mathbf{V}_{aug} \right) \hat{\mathbf{d}} \geq 0, \quad \forall \mathbf{d} \in \mathbb{R}^N_+. \tag{9}$$

This transformation successfully converts the intractable region-specific constraint into a standard copositive constraint on the matrix $\mathbf{Z} - \mathbf{E}_{11} - \lambda \mathbf{V}_{aug}$, which can be further approximated by semidefinite programming.

1.4. Tractable SDP Formulation

The reformulation above leads to a Copositive Programming (COP) problem, which remains intractable. To obtain a computable solution, we use the standard inner approximation of the copositive cone, decomposing a copositive matrix into the sum of a Positive Semidefinite (PSD) matrix and a non-negative matrix:

$$\mathcal{COP} \hookleftarrow \mathcal{S}_+ + \mathcal{N} = \{\mathbf{P} + \mathbf{N} \mid \mathbf{P} \succeq 0,\ \mathbf{N} \geq 0\}\,. \tag{10}$$

Replacing the copositive constraints with this approximation yields the following tractable Semidefinite Programming (SDP) formulation. Since the feasible set is restricted, the optimal value provides a safe upper bound on the worst-case violation probability:

$$\begin{aligned}
\min_{\substack{y_0,\mathbf{y},\mathbf{M},\lambda\\ \mathbf{P}_0,\mathbf{N}_0,\mathbf{P}_1,\mathbf{N}_1}} \quad & y_0 + \boldsymbol{\mu}^\top \mathbf{y} + \mathbf{Q} \bullet \mathbf{M} && (11)\\
\text{s.t.} \quad & \mathbf{Z} = \begin{pmatrix} y_0 & \frac{1}{2}\mathbf{y}^\top \\ \frac{1}{2}\mathbf{y} & \mathbf{M} \end{pmatrix}, && (12)\\
& \mathbf{Z} = \mathbf{P}_0 + \mathbf{N}_0, \quad \text{(Global Non-negativity)} && (13)\\
& \mathbf{Z} - \mathbf{E}_{11} - \lambda \mathbf{V}_{aug} = \mathbf{P}_1 + \mathbf{N}_1, \quad \text{(Violation Coverage)} && (14)\\
& \mathbf{P}_0, \mathbf{P}_1 \succeq 0, \quad \mathbf{N}_0, \mathbf{N}_1 \geq 0, && (15)\\
& \lambda \geq 0. && (16)
\end{aligned}$$

If the optimal value of this SDP is less than or equal to γ, the chance constraint is guaranteed to be satisfied.

2. Distributionally Robust Chance-Constrained Model

Scenario-based chance-constrained models approximate the true distribution using the empirical distribution constructed from historical samples. When the sample size is limited, such models may suffer from poor out-of-sample feasibility. To enhance robustness, we adopt *distributionally robust optimization* (DRO), where the true distribution is unknown but assumed to belong to an *ambiguity set* constructed from historical data. The chance constraints are then enforced under the worst-case distribution in this ambiguity set.

2.1. DRICC/DRJCC Districting Model

Let N be the set of basic units. Let c_{ij} be the cost of assigning unit i to district center j. Binary decision variable $x_{ij} = 1$ if unit i is assigned to district j and 0 otherwise. Let p be the number of districts. Let $\widetilde{\mathbf{d}} = [\widetilde{d}_1, \ldots, \widetilde{d}_N]^\top$ be the random demand vector.

The distributionally robust individual chance-constrained (DRICC) districting model is:

$$\min_{x_{ij}} \quad \sum_{i \in N} \sum_{j \in N} c_{ij} x_{ij} \tag{17}$$

$$\text{s.t.} \quad \sum_{j \in N} x_{ij} = 1, \qquad \forall i \in N, \tag{18}$$

$$\sum_{j \in N} x_{jj} = p, \tag{19}$$

$$\inf_{P \in \mathcal{D}} P\left\{ \frac{1-\alpha}{p} \widetilde{\mathbf{d}}^\top \mathbf{1} \ \le \ \widetilde{\mathbf{d}}^\top \mathbf{x}_j \ \le \ \frac{1+\alpha}{p} \widetilde{\mathbf{d}}^\top \mathbf{1} \right\} \ge 1 - \gamma, \quad \forall j \in N, \tag{20}$$

$$\sum_{i \in \cup_{v \in S}(N_v \setminus S)} x_{ij} - \sum_{i \in S} x_{ij} \ge 1 - |S|, \qquad \forall j \in N,\ S \subseteq [N \setminus (N_j \cup \{j\})], \tag{21}$$

$$x_{ij} \in \{0, 1\}, \qquad \forall i, j \in N. \tag{22}$$

Here (21) enforces contiguity via the standard cut-set formulation, where N_v denotes the neighbor set of node v in the adjacency graph. Constraint (20) represents the *distributionally robust relative balance requirement*, which ensures that, under the worst-case distribution in $\mathcal{D}$, the demand of each district remains within a tolerance level α around the average system demand.

If we require all districts to satisfy the relative balance constraints simultaneously with probability at least $1-\gamma$, we obtain the distributionally robust joint chance constraint (DRJCC):

$$\inf_{P\in\mathcal{D}} P\left\{ \frac{1-\alpha}{p}\,\widetilde{\mathbf{d}}^\top\mathbf{1} \;\le\; \widetilde{\mathbf{d}}^\top\mathbf{x}_j \;\le\; \frac{1+\alpha}{p}\,\widetilde{\mathbf{d}}^\top\mathbf{1}, \quad \forall j\in N \right\} \ge 1-\gamma. \tag{23}$$

Since DRJCC is typically much harder, we focus on DRICC first and then use a standard Bonferroni approximation for DRJCC (see Section 2.5).

2.2. Moment-Based Ambiguity Sets

We use moment-based ambiguity sets constructed from historical samples. Let $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ denote the sample mean and covariance of $\widetilde{\mathbf{d}}$, and let $\mathbf{Q}=\boldsymbol{\Sigma}+\boldsymbol{\mu}\boldsymbol{\mu}^\top$ be the second-moment matrix.

We consider two ambiguity sets:

$$\mathcal{D}_1 = \Big\{ P\in\mathcal{P}(\mathbb{R}^N_+):\ \mathbb{E}_P[\mathbf{d}]=\boldsymbol{\mu},\ \mathbb{E}_P[\mathbf{d}\mathbf{d}^\top]=\mathbf{Q} \Big\}, \tag{24}$$

$$\mathcal{D}_2 = \Big\{ P\in\mathcal{P}(\mathbb{R}^N_+):\ (\mathbb{E}_P[\mathbf{d}]-\boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1}(\mathbb{E}_P[\mathbf{d}]-\boldsymbol{\mu})\le\delta_1,\ \mathbb{E}_P[(\mathbf{d}-\boldsymbol{\mu})(\mathbf{d}-\boldsymbol{\mu})^\top]\preceq\delta_2\boldsymbol{\Sigma} \Big\}, \tag{25}$$

where $\delta_1>0$ and $\delta_2>1$ quantify estimation uncertainty.

2.3. Relative Balance DRICC

We focus on *relative* balance constraints. Let $\mathbf{1}$ be the all-ones vector. For each district j, define

$$\mathbf{x}_j=[x_{1j},\dots,x_{Nj}]^\top, \qquad \mathbf{v}_{j,L}=\frac{1-\alpha}{p}\mathbf{1}-\mathbf{x}_j, \qquad \mathbf{v}_{j,U}=\mathbf{x}_j-\frac{1+\alpha}{p}\mathbf{1},$$

where $\alpha\in[0,1)$ is the tolerance level. The relative balance DRICC for each district j is:

$$\inf_{P\in\mathcal{D}} P\Big\{ \widetilde{\mathbf{d}}^\top\mathbf{v}_{j,L}\le 0,\ \widetilde{\mathbf{d}}^\top\mathbf{v}_{j,U}\le 0 \Big\} \ge 1-\gamma, \quad \forall j. \tag{26}$$

Tractable DRICC Reformulations (Results Only).. To avoid repeating standard derivations, we directly state the tractable reformulations used in computation.

(i) SOC form under $\mathcal{D}_1$ (ADM-type approximation). Given a risk split $\gamma_a, \gamma_b \geq 0$ with $\gamma_a + \gamma_b = \gamma$, we enforce:

$$\boldsymbol{\mu}^\top \mathbf{v}_{j,L} + \sqrt{\frac{1-\gamma_a}{\gamma_a}}\ \sqrt{\mathbf{v}_{j,L}^\top \boldsymbol{\Sigma} \mathbf{v}_{j,L}} \leq 0, \quad \forall j, \tag{27}$$

$$\boldsymbol{\mu}^\top \mathbf{v}_{j,U} + \sqrt{\frac{1-\gamma_b}{\gamma_b}}\ \sqrt{\mathbf{v}_{j,U}^\top \boldsymbol{\Sigma} \mathbf{v}_{j,U}} \leq 0, \quad \forall j. \tag{28}$$

(ii) SOC form under $\mathcal{D}_2$ (ADM-type approximation). Let $\rho = \delta_1/\delta_2$. Depending on whether $\rho \leq \gamma_a$ (resp. γ_b), we use coefficients

$$t(\gamma) = \begin{cases} \sqrt{\delta_1} + \sqrt{\frac{1-\gamma}{\gamma}(\delta_2 - \delta_1)}, & \text{if } \rho \leq \gamma, \\ \sqrt{\frac{\delta_2}{\gamma}}, & \text{if } \rho > \gamma, \end{cases}$$

and enforce:

$$\boldsymbol{\mu}^\top \mathbf{v}_{j,L} + t(\gamma_a)\ \sqrt{\mathbf{v}_{j,L}^\top \boldsymbol{\Sigma} \mathbf{v}_{j,L}} \leq 0, \quad \forall j, \tag{29}$$

$$\boldsymbol{\mu}^\top \mathbf{v}_{j,U} + t(\gamma_b)\ \sqrt{\mathbf{v}_{j,U}^\top \boldsymbol{\Sigma} \mathbf{v}_{j,U}} \leq 0, \quad \forall j. \tag{30}$$

The above SOC constraints yield a MISOCP that can be solved directly or accelerated by outer approximation cuts (Section 2.4.1). In addition, for $\mathcal{D}_1$ we also consider an SDP safe approximation based on copositive programming (Section 2.4.2).

2.4. Solution Algorithms

2.4.1. (I) Supporting-Hyperplane Cut Algorithm for SOC Constraints

We treat each SOC constraint as a convex inequality $\varphi(\mathbf{x}) \leq 0$ in the continuous relaxation. If an incumbent $\bar{\mathbf{x}}$ violates it, we add a supporting-hyperplane cut:

$$\varphi(\bar{\mathbf{x}}) + \mathbf{g}^\top (\mathbf{x} - \bar{\mathbf{x}}) \leq 0, \qquad \mathbf{g} \in \partial\varphi(\bar{\mathbf{x}}). \tag{31}$$

Cuts are generated separately for (27)–(28) (or (29)–(30)). This yields a MILP-based outer approximation that iteratively tightens the SOC feasible region while retaining validity for binary x_{ij}.

2.4.2. (II) SDP Safe Approximation via Copositive Reformulation (Master–Subproblem)

This approach provides a *safe* (conservative) approximation for linear DR-CCs under $\mathcal{D}_1$ with support $\mathbb{R}_+^N$. Consider a generic linear DRCC:

$$\sup_{P \in \mathcal{D}_1} P(\mathbf{d}^\top \mathbf{v} > 0) \leq \gamma. \tag{32}$$

We use an SDP upper bound on the worst-case violation probability. Define decision variables $(y_0, \mathbf{y}, \mathbf{M})$ and

$$\mathbf{Z} = \begin{pmatrix} y_0 & \frac{1}{2}\mathbf{y}^\top \\ \frac{1}{2}\mathbf{y} & \mathbf{M} \end{pmatrix}, \qquad \mathbf{V}_{\text{aug}} = \begin{pmatrix} 0 & \frac{1}{2}\mathbf{v}^\top \\ \frac{1}{2}\mathbf{v} & \mathbf{0} \end{pmatrix}, \qquad \mathbf{E}_{11} = \text{matrix with } (1,1) = 1 \text{ and others } 0.$$

Using a standard copositive-to-SDP inner approximation $\mathcal{COP} \supseteq \mathcal{S}_+ + \mathcal{N}$, we solve:

$$\min_{\substack{y_0, \mathbf{y}, \mathbf{M}, \lambda \\ \mathbf{P}_0, \mathbf{N}_0, \mathbf{P}_1, \mathbf{N}_1}} \quad y_0 + \boldsymbol{\mu}^\top \mathbf{y} + \mathbf{Q} \bullet \mathbf{M} \tag{33}$$

$$\text{s.t.} \quad \mathbf{Z} = \mathbf{P}_0 + \mathbf{N}_0, \tag{34}$$

$$\mathbf{Z} - \mathbf{E}_{11} - \lambda \mathbf{V}_{\text{aug}} = \mathbf{P}_1 + \mathbf{N}_1, \tag{35}$$

$$\mathbf{P}_0, \mathbf{P}_1 \succeq 0, \quad \mathbf{N}_0, \mathbf{N}_1 \geq 0, \quad \lambda \geq 0. \tag{36}$$

The optimal value of (33)–(36) is a conservative estimate of $\sup_{P \in \mathcal{D}_1} P(\mathbf{d}^\top \mathbf{v} > 0)$. Thus, enforcing this value $\leq \gamma$ yields a safe approximation.

Master–Subproblem Implementation for Districting.. For the relative balance DRICC (26), we apply the SDP certificate separately to each linear form $\mathbf{v}_{j,L}$ and $\mathbf{v}_{j,U}$ (with a risk split γ_a, γ_b). Specifically, define the two violation events:

$$\mathcal{V}_{j,L} = \{\mathbf{d}^\top \mathbf{v}_{j,L} > 0\}, \qquad \mathcal{V}_{j,U} = \{\mathbf{d}^\top \mathbf{v}_{j,U} > 0\}.$$

A sufficient (safe) enforcement of (26) is:

$$\sup_{P\in\mathcal{D}_1} P(\mathcal{V}_{j,L}) \le \gamma_a, \qquad \sup_{P\in\mathcal{D}_1} P(\mathcal{V}_{j,U}) \le \gamma_b, \qquad \gamma_a+\gamma_b=\gamma, \quad \forall j. \tag{37}$$

Master problem. We solve the districting MILP with assignment, district-count, contiguity, and binary constraints:

$$\begin{aligned} \min_{x_{ij}} \quad & \sum_{i\in N}\sum_{j\in N} c_{ij}x_{ij} \\ \text{s.t.} \quad & (18)--(22), \end{aligned} \tag{38}$$

and we progressively add any violated robustness constraints detected by the subproblems (see below), either as linear cuts (if available) or by directly enforcing feasibility via certificate thresholds.

Subproblem(s). Given an incumbent $\bar{\mathbf{x}}$ from the master, for each district j we compute $\bar{\mathbf{v}}_{j,L}, \bar{\mathbf{v}}_{j,U}$ and solve the SDP certificate (33)–(36) twice:

- Subproblem-$L(j)$: set $\mathbf{v}=\bar{\mathbf{v}}_{j,L}$ and obtain an upper bound on $\sup_{P\in\mathcal{D}_1} P(\mathbf{d}^\top\bar{\mathbf{v}}_{j,L} > 0)$.
- Subproblem-$U(j)$: set $\mathbf{v}=\bar{\mathbf{v}}_{j,U}$ and obtain an upper bound on $\sup_{P\in\mathcal{D}_1} P(\mathbf{d}^\top\bar{\mathbf{v}}_{j,U} > 0)$.

If either bound exceeds γ_a or γ_b, the incumbent violates (37) and is rejected by adding the corresponding robustness constraint/cut back to the master. This yields a practical decomposition separating combinatorial decisions (master) and convex robustness verification (SDP subproblems).

2.5. DRJCC Reformulation via Bonferroni Approximation

For the joint chance constraint (23), we use the Bonferroni inequality. Let $\gamma_j \ge 0$ with $\sum_{j\in N}\gamma_j=\gamma$. If the DRICC constraints

$$\inf_{P\in\mathcal{D}} P\{\text{district } j \text{ satisfies the balance constraint}\} \ge 1-\gamma_j, \quad \forall j\in N,$$

hold, then the DRJCC is guaranteed:

$$\inf_{P \in \mathcal{D}} P\{\text{all districts satisfy the balance constraint}\} \geq 1 - \gamma.$$

In experiments we set $\gamma_j = \gamma/|N|$ for simplicity and tractability, and solve the resulting DRICC-based approximation using either SOC cuts (Section 2.4.1) or SDP safe approximation (Section 2.4.2).